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Java SE 27 & JDK 27

Class CallSite

java.lang.Object
    java.lang.invoke.CallSite

Direct Known Subclasses:

ConstantCallSite, MutableCallSite, VolatileCallSite

```
public abstract sealed class CallSite
extends Object
permits ConstantCallSite, MutableCallSite, VolatileCallSite
```

A `CallSite` is a holder for a variable `MethodHandle`, which is called its `target`. An `invokedynamic` instruction linked to a `CallSite` delegates all calls to the site's current target. A `CallSite` may be associated with several `invokedynamic` instructions, or it may be "free floating", associated with none. In any case, it may be invoked through an associated method handle called its dynamic invoker.

`CallSite` is an abstract sealed class which does not allow direct subclassing by users. It has three immediate, concrete non-sealed subclasses that may be either instantiated or subclassed.

- If a mutable target is not required, an `invokedynamic` instruction may be permanently bound by means of a constant call site.
- If a mutable target is required which has volatile variable semantics, because updates to the target must be immediately and reliably witnessed by other threads, a volatile call site may be used.
- Otherwise, if a mutable target is required, a mutable call site may be used.

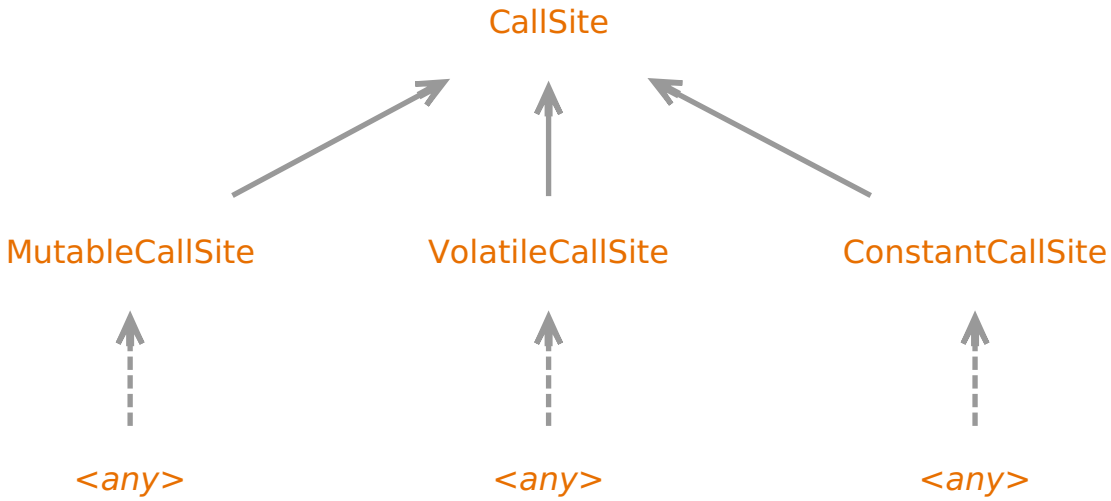
A non-constant call site may be *relinked* by changing its target. The new target must have the same type as the previous target. Thus, though a call site can be relinked to a series of successive targets, it cannot change its type.

Here is a sample use of call sites and bootstrap methods which links every dynamic call site to print its arguments:

```
static void test() throws Throwable {
    // THE FOLLOWING LINE IS PSEUDOCODE FOR A JVM INSTRUCTION
    InvokeDynamic[#bootstrapDynamic].baz("baz arg", 2, 3.14);
}
private static void printArgs(Object... args) {
  System.out.println(java.util.Arrays.deepToString(args));
}
private static final MethodHandle printArgs;
static {
```

```
  MethodHandles.Lookup lookup = MethodHandles.lookup();
  Class thisClass = lookup.lookupClass();  // (who am I?)
  printArgs = lookup.findStatic(thisClass,
      "printArgs", MethodType.methodType(void.class, Object[].class));
}
private static CallSite bootstrapDynamic(MethodHandles.Lookup caller, String
  // ignore caller and name, but match the type:
  return new ConstantCallSite(printArgs.asType(type));
}
```

Sealed Class Hierarchy Graph:



Since:

1.7

Method Summary

All Methods	Instance Methods	
Modifier and Type	**Method**	**Description**
abstract MethodHandle	**dynamicInvoker**()	Produces a method handle equivalent to an invokedynamic instruction which has been linked to this call site.
abstract MethodHandle	**getTarget**()	Returns the target method of the call site, according to the

		behavior defined by this call site's specific class.
abstract void	**setTarget**(MethodHandle newTarget)	Updates the target method of this call site, according to the behavior defined by this call site's specific class.
MethodType	**type**()	Returns the type of this call site's target.

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Method Details

type

```
public MethodType type()
```

Returns the type of this call site's target. Although targets may change, any call site's type is permanent, and can never change to an unequal type. The `setTarget` method enforces this invariant by refusing any new target that does not have the previous target's type.

Returns:

the type of the current target, which is also the type of any future target

getTarget

```
public abstract MethodHandle getTarget()
```

Returns the target method of the call site, according to the behavior defined by this call site's specific class. The immediate subclasses of `CallSite` document the class-specific behaviors of this method.

Returns:

the current linkage state of the call site, its target method handle

See Also:

`ConstantCallSite`, `VolatileCallSite`, `setTarget(MethodHandle)`, `ConstantCallSite.getTarget()`, `MutableCallSite.getTarget()`, `VolatileCallSite.getTarget()`

setTarget

```
public abstract void setTarget(MethodHandle newTarget)
```

Updates the target method of this call site, according to the behavior defined by this call site's specific class. The immediate subclasses of `CallSite` document the class-specific behaviors of this method.

The type of the new target must be equal to the type of the old target.

Parameters:

`newTarget` - the new target

Throws:

`NullPointerException` - if the proposed new target is null

`WrongMethodTypeException` - if the proposed new target has a method type that differs from the previous target

See Also:

`getTarget()`, `ConstantCallSite.setTarget(MethodHandle)`, `MutableCallSite.setTarget(MethodHandle)`, `VolatileCallSite.setTarget(MethodHandle)`

dynamicInvoker

```
public abstract MethodHandle dynamicInvoker()
```

Produces a method handle equivalent to an invokedynamic instruction which has been linked to this call site.

This method is equivalent to the following code:

```
MethodHandle getTarget, invoker, result;
getTarget = MethodHandles.publicLookup().bind(this, "getTarget", MethodType.
invoker = MethodHandles.exactInvoker(this.type());
result = MethodHandles.foldArguments(invoker, getTarget)
```

Returns:

a method handle which always invokes this call site's current target

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Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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